MLHVDC-LCS-ECI-DD-WF-0103-001A

30-B003701S-S0103-001

巴基斯坦默蒂亚里—拉合尔±660kV 直流输电工程
拉合尔±660kV 换流站工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

±660kV Lahore Converter Station

施工图设计
Detail Design

消 防 部 分

Fire Fighting Part

消防设计说明

Design Specification of Fire Fighting System

a) 中国电力技术装备有限公司
China Electric Power Equipment and Technology Co., Ltd.
b) 中国电力工程顾问集团华东电力设计院有限公司
East China Electric Power Design Institute of China Power Engineering Consulting Group
June, 2019

拉合尔±660kV 换流站 工程

施工图 阶段

中国电力工程顾问集团华东电力设计院有限公司

工程设计综合资质甲级 A131000025 工程勘察综合类甲级 B131000025

卷册检索号	30	B003701S	S0103
文本编号	B003701S	S0103	001

版号	日期	状态	修改内容摘要	设计	校核	审核	批准
0	201906	CFC	新制				

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任。

批准	2019.06	消防设计说明 Design Specification of Fire Fighting System
审核	2019.06	
	2019.06	
校核	2019.06	
设计	2019.06	

日期	2019.06	状态	CFC	版号	A

目录

1 概述 1
1.1 工程设计概况 1
1.2 设计范围 1
1.3 设计依据 1
1.4 初设收口审查意见 1
1.5 主要设计原则 2
1.6 采用的主要设计标准 2
2 消防给水 3
2.1 消防水量和消防水压 3
2.2 消防给水系统 3
2.2.1 消防水池 3
2.2.2 消防水泵房 4
2.2.3 室外消火栓系统 5
2.2.4 室内消火栓系统 5
2.2.5 水喷雾灭火系统 6
3 移动式灭火器 7

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

Catalog

1 Overview 9
1.1 Overview of Engineering Design 9
1.2 Design Scope 9
1.3 Design basis 9
1.4 Experts' review comments about The preliminary design document 10
1.5 Main Design Principles 10
1.6 Codes and Standards 11
2 Fire Fighting Supply 12
2.1 Fire Fighting Water Quantity & Hydraulic Pressure 12
2.2 Fire Fighting System 13
2.2.1 Fire Reservoir 13
2.2.2 Fire Pump House 13
2.2.3 Outdoor Hydrant System 15
2.2.4 Indoor Hydrant System 16
2.2.5 Water Spray System 16
3 Fire extinguisher 18

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

1 概述

1.1 工程设计概况

巴基斯坦默蒂亚里-拉合尔±660kV直流输电工程起点为信德省默蒂亚里，落点旁遮普省拉合尔，线路长度约878km，规划输电规模4000MW，由中国电力技术装备有限公司（CET）以BOOT模式开展。拉合尔换流站工程额定直流电压为±660kV，额定直流电流3030A，输送容量4000MW。

1.2 设计范围

本设计包括本工程范围内所有建（构）筑物、设备等的消防灭火系统，具体设计内容如下：

（1） 消防供水系统（消防泵房、消防水池）；

（2） 站区消防管网及室外消火栓灭火系统；

（3） 室内消火栓灭火系统；

（4） 水喷雾灭火系统；

（5） 移动式灭火器；

各专业防火措施及火灾报警控制系统详见相关专业技术描述及施工图。

注：（1）交流配电装置区 500kV 站用变消防设计不属于我院设计范围，详见中电装备院相关图纸。

（2）站前区办公楼、站前区中方宿舍、站前区中方食堂、站前区活动室、巴方宿舍、站前区巴方食堂、建维区办公楼、管理员宿舍、员工宿舍、巴方安保宿舍及站前区门卫室室内消防由我院委托都市设计院进行设计。

1.3 设计依据

（1）我院编制的本工程初步设计文件 30-B003701C-A01。

（2）本工程初步设计文件《初步设计说明书》审查会专家组意见。

（3）设计合同。

（4）有关单位来的设计输入资料。

1.4 初设收口审查意见

全站设置火灾探测报警系统，换流广场区域、站前区和建维区设室外消火栓系统；控制楼、办公楼、检修备品库等设室内消火栓系统；阀厅室内设移动式化学灭火装置；其他建筑物采用化学灭火器灭火；换流变压器、500kV 降压变压器消防均采用水喷雾灭火系统。

执行情况：

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

1、按审查意见，本工程在换流广场区域、站前区和建维区设室外消火栓系统。

2、按审查意见，控制楼、办公楼、检修备品库等设室内消火栓系统。

3、按审查意见，阀厅设室内设移动式化学灭火装置。

4、按审查意见，其他建筑物采用化学灭火器灭火。

5、按审查意见，换流变压器消防采用水喷雾灭火系统。

1.5 主要设计原则

(1) 本工程消防系统的设计，遵照国家“预防为主、防消结合”的方针，并根据工程具体情况，力求体现当前的消防设计思想和水平。从全局出发，统筹兼顾，正确处理生产和安全、重点和一般的关系，做到促进生产，保障安全，方便使用，经济合理。

(2) 消防系统在火灾初期发出报警信号，能进行火灾的集中、区域、就地监控和消防装置的远方和就地控制，并设有火灾一旦发生就足以扑灭的设备容量，该容量按同一时间内火灾数一次设计。

(3) 本工程消防采用设消防气压稳压装置的临时高压消防给水系统。

(4) 在站前区设消防水池及消防泵房。消防水泵房内设 2 台 50％电动消防泵和 1 台 100％柴油驱动消防泵及 1 套气压稳压装置，供全站水消防用水。

(5) 根据相关规范要求，本工程换流变压器设置水喷雾灭火系统；在设计范围内配置完善的火灾探测报警系统和移动式灭火器。

(6) 本工程消防灭火提升措施仅将消防水池与工业水池通过应急补水泵连接，事故时从工业水池应急补水，相关设计详见我院水布专业图纸。

1.6 采用的主要设计标准

《火力发电厂与变电站设计防火规范》	GB 50229-2006
《消防给水及消火栓系统技术规范》	GB 50974-2014
《建筑设计防火规范》	GB 50016-2014（2018 年版）
《水喷雾灭火系统技术规范》	GB 50219-2014
《火灾自动报警系统设计规范》	GB 50116-2013
《建筑灭火器配置设计规范》	GB 50140-2005
《电力设备典型消防规程》	DL5027-2015
《建筑机电工程抗震设计规范》	GB50981-2014

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

以上规范和标准均采用最新版本。

2 消防给水

2.1 消防水量和消防水压

根据施工图最终计算，站区主要建（构）筑物消防用水量和水压的计算参见下表 2.1-1 和 2.1-2。

表 2.1-1 换流站消防用水量表

序号	消防对象		消防水量 (l/s)	火灾延续时间 (min)	消防用水总量 (m^3)
1	主控楼	室外消火栓	15	120	180
		室内消火栓	10		
2	换流变	室外消火栓	15	120	394.0
		水喷雾	198.6	24	
3	站前区办公楼	室外消火栓	25	120	288
		室内消火栓	15		
	最大一次消防水量		213.6		394.0

表 2.1-2 换流站消防水压计算表

序号	消防设施所需水柱	换流变	主控楼	站前区办公楼
1	最不利点几何扬程(m)	13	14	9
2	水枪出口需要水头		35	35
3	喷头出口需要水头	35	0	0
4	管网损失	38.9	12.5	15.5
	合计总水头	86.9	61.5	59.5

根据上表可知，本工程消防最大用水流量为换流变所需 213.6l/s(769.0m^3/h)，最大水头为换流变水喷雾所需 0.869MPa，火灾发生时所需最大一次消防用水总量为换流变水喷雾以及室外消火栓用水量所需 394.0m^3。所设消防水泵流量、扬程及消防水池有效容积均满足要求。

2.2 消防给水系统

2.2.1 消防水池

本工程新建 2 座 300m^3 消防水池，位于站前区。水池采用半地下式钢筋混凝土结构，单

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

座水池内壁平面尺寸 B×L 约为 9.9m×9.9m，净高 3.5m，消防贮水量为 600 m^3。每座水池补水管为一路 DN200 进水管，并能保证补水时间不大于 48h。消防水池水源自站外深井泵房来。

根据业主要求，本工程将消防水池与工业水池通过应急补水泵连接，发生重大火灾时可从工业水池应急补水，作为消防提升措施，相关设计详见我院水布专业图纸。

消防水池布置及水池中关于消防的水位及报警信号详见 30-B003701S-S0501 卷册。

2.2.2 消防水泵房

本工程新建一座消防水泵房，位于站前区。消防泵房为半地下式布置，平面尺寸为 26.5m×10m，地下部分深 2.80m，地面部分高约 8.50m。泵房内设电动单梁悬挂桥式起重机一套，并设有检修场地，以便检修。

泵房内设置排水沟，沟末端设置集水坑，坑内设排水泵 2 台。一台参数为 Q=50m^3/h，H=10m，N=3kW，另一台参数为 Q=5m^3/h，H=8m，N=0.75kW。排水泵根据水位信号启停。泵停止、运行、故障、报警信号送至控制室，排水泵自带就地控制柜和浮子开关。

消防泵房内设两台 50%电动消防泵，一台 100%柴油机备用消防泵，一套消防气压稳压装置。设备参数为：电动消防泵 Q=450m^3/h，H=0.90MPa，N=200kW；柴油机备用消防泵 Q=800m^3/h，H=0.90MPa，N=351kW；气压稳压装置的气压罐有效容积为 3.6m^3，配 2 台补水泵（一用一备），Q=18m^3/h，H=0.97MPa，N=11kW。

消防泵房内手动按钮可启动、停止电动消防泵、柴油机消防泵；当电动消防泵启动失败自动启动柴油机消防泵；主消防盘（位于主控楼内）手动按钮可启动、停止电动消防泵、柴油机消防泵。消防控制柜或主消防盘应设置专用模块通过通讯电缆或光缆与手动启泵按钮连接。

气压稳压设备维持全站消防供水系统管网压力，使之经常处于 0.97-1.02MPa（可调）；当消防供水管网压力下降至 0.90MPa（可调）时，自控柜向#1 电动消防泵发出启动电泵的信号；当供水管网压力下降至 0.85MPa（可调）时，自控柜向#2 电动消防泵发出启动电泵的信号；当供水管网压力下降至 0.80MPa（可调）或电动消防泵发出故障信号时，立即直启动备用柴油消防泵。具体压力值在现场安装调试及运行时最终确定。

气压稳压装置为自动化控制，一旦补水泵故障，即发出声光报警信号，同时备用补水泵切换自动投入。气压稳压设备自控柜应就地显示补水泵的启、停、正常运转、故障及系统失电信号，信号并应送至站区主消防盘。

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

消防控制柜上可选择电动消防泵、柴油机消防泵"自动"或"非自动"运行状态，并留有显示，站区主消防盘亦留有显示。平时应使消防水泵处于自动启泵状态。停泵应由具有管理权限的工作人员根据火灾扑救情况确定后人工停泵。控制柜应就地显示消防水泵的启、停、正常运转、故障(包括柴油机泵综合故障)及系统失电信号，信号并应送至站区主消防盘。并接受消防主盘的启动命令。

控制柜上能够显示消防水池的水位，能够显示消防水池高水位和低水位声光报警信号。报警信号应送至站区主消防盘。

消防泵房设控制盘就地控制消防泵和稳压泵启停,消防泵也可以在主控楼消防总盘上遥控启停，并应有保护装置。

消防水泵控制柜应设置机械应急启泵功能，并应保证在控制柜内的控制线路发生故障时由有管理权限的人员在紧急时启动消防水泵。机械应急启动时，应确保消防水泵在报警后 5min 内正常工作。

有关消防泵房内设备的安装及产品特性详见 30-B003701S-S0502 卷册。

2.2.3 室外消火栓系统

本工程阀厅\换流变区域周围的消防环管管径采用 DN350mm，其他区域的消防水管管径为 DN200mm。沿换流站主控楼、阀厅、换流变区域、站前区、建维区周围道路环状布置，按照规范要求布置室外消火栓。

室外消火栓均由站区消防水管网上接出，工艺布置区的环形管网上，消火栓间距一般不超过 60m。站前区及建维区的环形管网上，消火栓间距一般不超过 80m。室外消火栓采用 DN100mm 防撞型室外地上式消火栓（SSF100/65-1.6），配置 120m（6×20m）长麻质衬胶水龙带、直流喷雾两用水枪各 1 套。消防水龙带、水枪平时可分散均置在站区各构筑物或就地消防小室内，便于扑救火灾时方便取用。

站区消防水管网上设有检修阀门，由阀门将环管分成若干段，每段内室外消火栓的数量不超过 5 个。

室外消防水管采用钢管（内外热浸镀锌），具体施工安装及防腐等要求详见 30-B003701S-S0503 卷册。

2.2.4 室内消火栓系统

本工程在主控楼、检修备品库、站前区办公楼、站前区中方宿舍、巴方宿舍、建维区办

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

公楼、管理人员宿舍楼、员工宿舍楼及巴方安保宿舍楼等建筑物内设置室内消火栓。其室内消火栓布置分别详见相应建筑物室内消火栓卷册。

各建筑物自室外消防水环管上引支管进入室内，在走廊、楼梯口等处设置室内消火栓，保证有两只水枪的充实水柱同时到达室内任何部位。室内消火栓采用减压稳压消火栓。室内消火栓设置在明显且易于取用的地点，栓口距地面高度为 1.1m。

消防柜采用 1800×700×240mm 单栓带灭火器箱组合式消防柜（乙型）（包括消防柜、消火栓、阀门、水枪、水带、消防软管卷盘、消防按钮等），并在每个消防柜内放置手提式干粉（磷酸铵盐）灭火器 2 具。每个消火栓处设消防报警按钮，报警信号接至火灾探测报警系统。消火栓采用单出口消火栓，栓口及水龙带直径为 DN65mm，水龙带长 25m，配 D19mm 直流/喷雾两用水枪；消防软管卷盘应配置内径不小于∅19mm 的消防软管,长度为 30m。

2.2.5 水喷雾灭火系统

（1） 换流变压器水喷雾灭火系统

本工程共有 12 台换流变压器，均采用水喷雾灭火系统。全站共设 4 座雨淋阀箱，均布在换流变附近，每座雨淋阀箱内设置 3 套雨淋阀组，分别对应 3 台换流变压器的水喷雾灭火系统。水源接自站区消防水环网，分别引两路 DN300mm 的消防水管接至 4 座雨淋阀室。换流变压器水喷雾灭火系统采用 DN250mm 的雨淋阀组，共 12 套。换流变压器本体及油枕水喷雾灭火系统的设计强度为：20L/min.m^2，油坑水喷雾灭火系统的设计强度为：6L/min.m^2。事故时消防排水和换流变压器油排入水布专业负责设计的变压器事故油池。

每套固定式水喷雾设施均由雨淋阀组、配水管网、高速水雾喷头、火灾探测控制系统组成，具有自动/手动喷水灭火保护，并报告火灾危险和反馈报警阀组状态等功能。其中雨淋阀组除报警阀及配套控制、测试、排水管路外，还包括检修信号阀、过滤器、控制电磁阀、报警压力开关、水力警铃、压力表等配套设备，具有消防控制、报警和测试功能。

水喷雾灭火系统设有自动控制、手动控制和应急操作三种控制方式。火灾探测器类型选用缆式线型感温，详见电气专业图纸。在发生火灾，探测器信号动作后，可自动启用雨淋阀组进行水喷雾灭火，也可采用人工手动控制及应急操作的方式。换流变压器水喷雾灭火系统的操作，除上述自动、手动、紧急启动及手、自动选择等功能外，还与换流变压器

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

保护系统联动，当火灾探测系统接到换流变压器保护系统的要求启动相应换流变压器水喷雾灭火系统的信号时手动启动该水喷雾灭火系统，或当火灾探测器动作和换流变压器跳闸同时发生时自动启动相应换流变压器水喷雾灭火系统。在雨淋阀本体上设有紧急机械启动装置，应采取安全保护措施，由专人负责，以免发生误操作，造成不必要的损失。

关于换流变压器水喷雾灭火系统的施工安装详见 30-B003701S-S0505 卷册。

（2）柴油消防泵水喷雾灭火系统

消防泵房内柴油消防泵及油箱采用水喷雾灭火系统。水源接自消防泵出水母管，设一套 DN150mm 雨淋阀组。其水喷雾灭火系统的设计强度均为：20L/min.m^2。消防泵房为半地下室，其消防排水由泵房内设置的排水沟排入集水坑，再通过坑内的潜水排污泵排至附近雨水窨井。

水喷雾灭火系统设有自动控制、手动控制和应急操作三种控制方式。火灾探测器类型选用缆式线型感温，详见电气专业图纸。在发生火灾，探测器信号动作后，可自动启用雨淋阀组进行水喷雾灭火，也可采用人工手动控制及应急操作的方式。在雨淋阀本体上设有紧急机械启动装置，应采取安全保护措施，由专人负责，以免发生误操作，造成不必要的损失。

关于柴油消防泵水喷雾灭火系统的施工安装详见 30-B003701S-S0506 卷册。

交流配电装置区 500kV 站用变消防灭火设计由中电装备院负责设计，不在我院设计范围内。

3 移动式灭火器

对站区的各个建筑物，均按《建筑灭火器配置设计规范》(GB50140－2005)、《电力设备典型消防规程》DL5027-2015 及《火力发电厂与变电站设计防火规范》(GB50229－2006)的要求，分别设置了手提式移动式灭火设施、推车式灭火器、砂箱等移动式灭火设施。灭火器的类型有干粉灭火器等.

灭火器设置在明显和便于取用的地方，且不得影响安全疏散，灭火器不应设置在潮湿或强腐蚀性的地方。手提式移动式灭火器可与室内消火栓箱一起放置在带灭火器箱组合式消防柜内，也可单独设置在移动式灭火器箱，可根据现场情况确定。灭火器箱可根据业主要求进行编号。一个灭火器配置场所内的灭火器不少于 2 具。露天布置的消防器材应有遮阳挡水、保温隔热的措施（由灭火器厂家配套提供）。

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

本工程主控楼、检修备品库、站前区办公楼、站前区中方宿舍、站前区中方食堂、站前区活动室、巴方宿舍、站前区巴方食堂、建维区办公楼、管理人员宿舍楼、员工宿舍楼、巴方安保宿舍楼、门卫室的移动式灭火器布置分别见其室内消火栓和灭火器布置图。其余建筑物的移动式灭火器布置及注意事项详见本工程 30-B003701S-S0519 卷册。

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

1 Overview

1.1 Overview of Engineering Design

The starting point of Pakistan's Mertiary-Lahore (+660 kV) HVDC transmission project is from Mertiary in Sindh Province to Lahore in Punjab Province, with a transmission line length of about 878 km and a planned transmission scale of 4000MW. The project is carried out by China Electric Power Technology Equipment Co., Ltd (CET) in the BOOT mode. The rated DC voltage is ±660kV, rated DC current is 3030A, and transmission capacity is 4000MW.

1.2 Design Scope

The design scope includes the fire fighting system of all the buildings and structures. Detail design scope is as follows.

（1） Fire fighting water supply system(fire pump house, fire reservoir);

（2） Outdoor fire hydrants & pipe network;

（3） Indoor fire hydrant system;

（4） Water spray system;

（5） Fire extinguisher.

Fire prevention measures of all departments and fire alarm control system refer to instructions and drawings of other departments.

Notes: (1) The fire fighting design of 500kV station transformers in AC distribution device area refers to relates drawings of China Electric Power Equipment Institute.

(2) The indoor fire fighting design of Office Building(The First Building Area), Dormitory Building for Chinese, Refectory for Chinese, Activity House, Dormitory Building for Pakistani, Refectory for Pakistani, Guard Building, Office building (Temporary Building Area), Dormitory Building for Managers, Dormitory Building for Staff, Dormitory Building for Security Personnel has been entrusted to Urban Design Institute.

1.3 Design basis

（1）The preliminary design document(30-B003701C-A01) of Lahore (+660 kV) converter station project;

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

（2）Experts' review comments about the preliminary design document;

（3）Design contract;

（4）Input design data of related companies。

1.4 Experts' review comments about The preliminary design document

Fire alarm control system should be set all over the station; the outdoor fire hydrants & pipe network of this project should be set all over the station except DC field, AC field and AC filter field; Indoor fire hydrant system should be set in control building, office building, maintenance workshop & spare parts storage and etc; Mobile chemical fire extinguishing device should be set in valve halls and maintenance workshop & spare parts storage; Chemical fire extinguisher should be set in other buildings; Converter transformers and 500kV station transformers should be equipped with water spray system. Fire fighting design should be according to Chinese codes, relevant explanation should be supplied.

Implementation:

1. The outdoor fire hydrants & pipe network of this project are set all over the station except DC field, AC field and AC filter field according to review comments.

2. Indoor fire hydrant system are set in control building, office building and maintenance workshop & spare parts storage according to review comments.

3. Chemical fire extinguishing device are set in valve halls and maintenance workshop & spare parts storage according to review comments.

4. Chemical fire extinguisher are set in other buildings according to review comments.

5. Converter transformers are equipped with water spray system according to review comments.

1.5 Main Design Principles

(1) The design of fire fighting system should follow the guideline "prevention first, combine prevention with elimination" and the specific conditions of the project, that strive to reflect the current fire protection design thought and level. Reduce unnecessary spare, and reduce

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

investment as far as possible.

(2) Fire alarm signals are sent out in the early stage of a fire, which can be used for centralized, regional, on-site monitoring and remote and local control of fire fighting devices. The capacity of the equipment is designed according to a fire at the same time.

(3) The fire fighting system of this project adopts fire water supply system of temporary high pressure with pressure stabilizer.

(4) Fire pump house and fire reservoir are set in the first building area. Two 50% electric fire pumps, one 100% diesel fire pump and a set of pressure stabilizer are set in the fire pump house, supplying fire water for the whole fire fighting system.

(5) According to relevant codes, converter transformers are equipped with water spray system; fire alarm control system and fire extinguishers are set in the design scope.

(6) The promotion measure required by general contractor is as follow: connecting industrial reservoir and fire reservoir with emergency water supply pump in case of fire, and relevant design refer to drawings of hydraulic layout department.

1.6 Codes and Standards

Code for design of fire protection for fossil fuel power plants and substations
(GB 50229-2006)

Technical code for fire protection water supply and hydrant systems
(GB50974-2014) (2018 edition)

Code for fire protection design of buildings (GB50016-2014)

Technical code for water spray fire protection systems (GB50219-2014)

Code for design of automatic fire alarm system (GB50116-2013)

Code for design of extinguisher distribution in buildings (GB50140-2005)

Typical extinguishing and protection regulation of electrical equipment
(DL5027-2015)

Code for seismic design of mechanical and electrical equipment
(GB50981-2014)

Notes: latest version of codes above should be adopted.

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

2 Fire Fighting Supply

2.1 Fire Fighting Water Quantity & Hydraulic Pressure

According to the final calculation, the fire fighting water quantity of main buildings of the station is shown in table 2.1-1, and the worst-case fire fighting water pressure calculation is shown in table 2.1-2.

Table 2.1-1 Fire fighting water quantity of main buildings

No.	Fire fighting object		Fire fighting water flow（L/s）	Fire duration (min)	fire fighting water quantity (m^3)
1	Control Building	Outdoor hydrant	15	120	180
		Indoor hydrant	10		
2	Convertor Transformer	Outdoor hydrant	15	120	394.0
		Water spray	198.6	24	
3	Office Building(The First Building Area)	Outdoor hydrant	25	120	288
		Indoor hydrant	15		
	Maximum Fire Fighting Water Quantity		213.6		394.0

Table 2.1-2 The worst-case fire fighting water pressure calculation

No.	Pressure required for Fire Installations（m）	Convertor Transformer	Control Building	Office Building(The First Building Area)
1	Height of the most unfavorable point(m)	13	14	9
2	Pressure required for hydrant head(m)		35	35

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

3	Pressure required for sprinkler head(m)	35	0	0
4	Hydraulic loss of pipe network(m)	38.9	12.5	15.5
	Total pressure required (m)	86.9	61.5	59.5

According to the above tables, the maximum fire fighting water flow of this project is 213.6l/s(769.0m^3/h) for Convertor Transformer; the maximum total pressure required is 0.869MPa for Convertor Transformer; the maximum fire fighting water quantity is 394.0m^3 for Convertor Transformer. The fire pump and fire reservoir of the project all can meet the requirements.

2.2 Fire Fighting System

2.2.1 Fire Reservoir

Two 300m^3 fire reservoirs are newly built in the first building area. The structure of the tank is half buried structure. Size of inner wall of single reservoir is 9.9m×9.9m, and height is 3.5m. Fire water quantity is 600m^3. Each reservoir set one DN200 inlet pipe, which can ensure replenishing time within 48h. Water is from the deep well pump house water supply pipe outside the substation.

The promotion measure required by general contractor is as follow: connecting industrial reservoir and fire reservoir with emergency water supply pump in case of fire, and relevant design refer to drawings of hydraulic layout department.

The arrangement of fire reservoir and alarm level condition refers to 30-B003701S-S0501.

2.2.2 Fire Pump House

Fire pump house is newly built in the first building area. The structure of the pump house is half buried structure. Size of inner wall is 26.5m×10m, depth 2.80m, height about 8.50m. One set of electric crane has been equipped in pump house for overhaul.

Drainage ditch has been set in pump house, the end of which is collection sump, and 2 drainage pump set in the sump. The parameter of one is Q=50m^3/h, H=10m, N=3kW, while the

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

other is Q=5m^3/h, H=8m, N=0.75kW. Starting and stopping the pump according to water level. The signal of pump stop, operation, fault, and alarm signals must be sent to the control room or fire alarm panel. The drainage pumps are equipped with a local control cabinet and a float switch.

Two 50% capacity of electric motor fire pumps, a 100% capacity of diesel fire pump and a set of pressure stabilizing device has been set in fire pump house. The parameter is as follow: electric motor fire pump: Q=450m^3/h, H=0.90MPa, N=200kW; diesel fire pump: Q=800m^3/h, H=0.90MPa, N=351kW; effective volume of pressure tank: 3.6m^3; two pressure stabilizing pumps, one-up: Q=18m^3/h, H=0.97MPa, N=11kW.

The motor fire pump and diesel fire pump can be activated and shut down manually down manually by main fire protection panels in control building. The fire control cabinet in fire pump house. The diesel fire pump is activated automatically when motor fire or the main fire protection panels should be connected with manual pump start button pump start failed. The motor fire pump and diesel fire pump can be activated and shut by special module and the communication cable.

The pressure of firewater piping network at stable pressure of about 0.97~1.02MPa(adjustable) is maintained by firefighting pressure stabilizing device. The activated signal is sent to #1 motor fire pump by control cabinet when the pressure of fire water piping drops to 0.90Mpa(adjustable).The activated signal is sent to #2 motor fire pump by control cabinet when the pressure of fire water piping drops to 0.85Mpa(adjustable).When the piping pressure continues to drop to 0.80Mpa(adjustable) or the malfunction signal of fire motor pump is sent, the diesel fire pump is activated.

The pressure stabilizer is of automation control. When pressure stabilizing pump is in fault, the detector will send out sound and light alarm signal, and the sparing one used automatically. The control cabinet has the function of displaying the signal of on, off, normal operation, fault and out of battery, and the signal should be sent to the main control panel.

The fire control cabinet has the function of choosing the working status electric motor fire pump and diesel fire pump automatic or not. The pump should be in automatic status in normal time, and stopping the pump should be in charge of staff with management authority according to

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

the fire condition. The control cabinet has the function of displaying the signal of on, off, normal operation, fault and out of battery, and the signal should be sent to the main control panel. And it can accept the signal from the main control panel.

The elevation of the fire reservoir can be showed on the control cabinet, and the sound and light alarm signals of fire reservoir high and low water level can be showed on the control cabinet also. The alarming signal should be sent to the fire protection panels in control building.

Local control panel is installed in fire pump house to control fire water pumps which can also be controlled from main control panel in the control room, and the local control panel should set self-protection device.

The fire pump cabinet has the function of emergency start-up. When in the case of emergency start-up, the pump should start in 5min since alarming.

The installation and product characteristics refer to 30-B003701S-S0502.

2.2.3 Outdoor Hydrant System

The diameter of the fire pipe network around the valve hall & convertor transformers yard is DN350mm, and the fire pipe network of other areas is DN200mm. It is looped in the control building, valve hall, transformers field, first building area and temporary building area, and outdoor fire hydrants are set according to codes.

Outdoor fire hydrants extend from the fire pipe network. The distance between fire hydrants is generally no more than 60m on the ring pipe network in the process unit areas. The distance between fire hydrants is generally no more than 80m in the ring pipe network of the operation & maintenance area and temporary building area. Outdoor hydrants adopt DN100mm anti-collision outdoor hydrants (SSF100/65-1.6). Each outdoor fire hydrant is configured with 1 set of 120m (6*20m) long hemp lining dragon belt with inner liner, 1 set of flow/spray dual-use squirt gun 1 sets. The fire hose and the squirt guns can be arranged in the buildings and the fire chambers factory in the yard, which is convenient to use.

Repair valve has been set in the fire pipe network, dividing the network to several parts, ensuring that the hydrants number is within 5 in each part.

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

The outer buried fire pipe of the project adopts thickened inside and outside hot-dip galvanized steel pipes.

The construction requirements and details refer to 30-B003701S-S0503.

2.2.4 Indoor Hydrant System

Indoor hydrant system has been set in the building of this project as follow: The Control Building, Maintenance Workshop & Spare Parts Storage, Office Building(The First Building Area), Dormitory Building for Chinese, Dormitory Building for Pakistani, Office building (Temporary Building Area), Dormitory Building for Managers, Dormitory Building for Staff and Dormitory Building for Security Personnel. The indoor hydrant arrangement refer to relevant drawing.

Building with indoor hydrant system extend fire pipe from the fire pipe network, setting hydrant in the position of corridor, stairs and etc, ensuring that there are two hydrants can reach any part. The indoor fire hydrant in this volume shall use the type pressure reducing and stabilizing hydrant. It should be set in those places obvious and easy to get. The hydrant head should be 1.1m height to the floor.

The fire hydrant box adopts the model of single bolt and fire extinguisher box combined fire hydrant box(type B), the size of the fire hydrant box is 1800*700*240mm. (The fire hydrant box including fire cabinets, fire hydrant, valves, water guns, hoses, fire hose reel, fire button, etc.), and two portable dry powder fire extinguishers are placed in each fire hydrant box. Each fire hydrant is equipped with fire alarm button, alarm signals are connected to the fire detection and alarm system. Fire hydrant with single outlet is set D19mm DC/spray water gun. The diameter of hydrant outlet is DN65mm, water hose length is 25m.Fire hose reel should be equipped with fire hose diameter is not less than 19mm of the length of up to 30m.

2.2.5 Water Spray System

(1) Converter Transformer Water Spray System

There are 12 converter transformers in this project, all equipped with water spray system. There are 4 deluge valve boxes, setting near the converter transformers. 3 deluge valves are set in

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

one box. Each box extends two DN300mm fire pipe from the fire pipe network. The converter transformer water spray system adopts DN250mm deluge valve, 12 sets in all. The design strength of the water spray system of the converter transformer, oil station and fan is 20L/min.m^2.The design strength of water spray system for oil pit is 6L/min.m^2. Fire water and the oil of transformer in case of fire drain into accident oil pool, the design of which refer to drawing of hydraulic layout department.

Each water spray fire extinguishing system is composed of deluge valve, fire pipe, high speed water spray nozzle and fire detection and control system, and has the function of automation or not, reporting fire situation and valve status. And the deluge valve contains valve, controlling, testing and draining pipe, maintenance signal valve, filter, control solenoid valve, alarm pressure switch, hydraulic alarm bell, pressure gauge and other corollary equipment, and with the function of fire controlling, alarming and testing.

Water spray fire extinguishing system should be equipped with three kinds of control methods: automatic control, manual control and emergency operation. The type of fire detector uses cable type temperature sensing, and it is detailed in electrical drawings. After fire and detector signal action, it can automatically activate the deluge valves for water spray fire extinguishing, or manual control and emergency operation. When used for transformer water spray extinguishing system operation, in addition to the automatic, manual, and emergency start hands, automatic selection and other functions, the system should interact with the transformer protection system. When the signal fire detection system receives system of transformer protection requirements of starting corresponding transformer water spray extinguishing system，the spray system should be started manually, or when the fire detector action and the transformer trip occur at the same time, the corresponding transformer water spray fire extinguishing system is automatically started. An emergency mechanical starting device shall be installed on the body of the shower valve, and safety protection measures shall be taken by the special person responsible for it, so as to avoid misoperation and cause unnecessary losses.

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

Detail of the converter transformer water spray system refers to 30-B003701S-S0505.

（2）Diesel Pump Water Spray System

The diesel pump and diesel tank in the fire pump house is equipped with Water Spray System, extending one DN150 fire pipe from the main pipes of the fire pipe network, adopting one DN150 deluge valve. The design strength of the water spray system of the Diesel Fire Pump and diesel tank is 20L/min.m^2. The structure of the pump house is half buried structure, and fire water drains to the collection sump through drainage ditch, and drain to rainwater well nearby by drainage pump.

Water spray fire extinguishing system should be equipped with three kinds of control methods: automatic control, manual control and emergency operation. The type of fire detector uses cable type temperature sensing, and it is detailed in electrical drawings. After fire and detector signal action, it can automatically activate the deluge valves for water spray fire extinguishing, or manual control and emergency operation. An emergency mechanical starting device shall be installed on the body of the shower valve, and safety protection measures shall be taken by the special person responsible for it, so as to avoid misoperation and cause unnecessary losses.

Detail of the diesel pump water spray system refers to 30-B003701S-S0506.

The fire fighting design of 500kV station transformers in AC distribution device area refers to relates drawings of China Electric Power Equipment Institute.

3 Fire extinguisher

Fire extinguisher facilities like portable fire extinguishers, wheeled fire extinguisher fire bucket and so on are set in each building of the substation according to GB50140-2005, DL5027-2015 and GB50229-2006. Mobile fire extinguishers should be located in obvious location, convenient for use and no impact on safe evacuation and should not be located in place dump and with strong corrosiveness. Portable fire extinguishers can be set with hydrant or alone in fire extinguisher box. Fire extinguisher can be identified according to the requirement of owner. At least two fire extinguishers should be set in one place. The fire fighting equipment (Fire extinguisher、 sand box and so on) should be equipped with the protection of the water and high temperature(Provided by the fire extinguisher manufacturer).

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

The fire extinguisher arrangement of control building, maintenance workshop & spare parts storage, office building(the first building area), dormitory building for Chinese, refectory for Chinese, activity house, dormitory building for Pakistani, refectory for Pakistani, office building(temporary building area), dormitory building for managers, dormitory building for staff, dormitory building for security personnel and guard room refer to relevant drawing. And the detail of other building fire extinguishers refers to 30-B003701S-S0519.

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任